Decennial Census Return Rates: The Role of Social Capital

by

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Abstract

This paper explores how useful information about social and civic engagement (social capital) might be to the U.S. Census Bureau in their efforts to improve predictions of mail return rates for the Decennial Census (DC) at the census tract level. Through construction of Hard-to-count (HRC) scores and multivariate analysis, we find that if information about social capital were available, predictions of response rates would be marginally improved.

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1 Introduction and Background

Article 1, Section 2 of the U.S. Constitution directs that an "enumeration" of the population be made every ten years; the first Decennial Census (DC) was taken in 1790. As the population of the U.S. and the uses to which this enumeration has been put has grown, so has the cost -- from $44,000 (1.1 cents per capita) in 1790 to $12.9 billion ($48 per capita) in 2010 (U.S. Census Bureau 2016).

One of the most challenging aspects of conducting the DC is connecting with "Hard-to-Count" (HTC) members of the population, including households that did not return their questionnaire by mail. For the 2010 DC, the Census Bureau undertook new and innovative measures to reach the HTC. In spite of these efforts, and additional expense (109% increase in per capita cost over 2000), the Government Accounting Office (GAO) reported that the main return rates in 2010 were the same as in the 2000 DC (GAO 2010).[1] The same report recommends that the Census Bureau undertake the development of a predictive model to better anticipate those areas with HTC populations in order to target resources and allocate staff across geographic areas effectively and efficiently (GAO 2010, 32).

The purpose of this paper is to assess whether additional information about a community's level of social capital can help to better predict a return rate from that area, in order to better target resources to improve mail-in responses.[2] Two key factors are necessary for such a

[1] With the growth in diversity and other socioeconomic trends, this non-decline in return rates was considered a positive accomplishment.

[2] "Mail **return** rates are based on occupied housing units, while the denominator of the mail **response** rates include vacant as well as occupied units. So a return rate will be higher than a response rate, and more closely reflect the participation of households in the census (a vacant unit cannot return a form)," [emphasis added] (Bruce and Robinson undated, 13).

strategy to be successful. First of all, there needs to be a positive connection between a person's (or area's) level of social capital and their probability of responding to surveys, especially the DC. Second, there needs to be some evidence that additional expenditures, usually through use of targeted media, is successful in improving the probability of responding. There is some evidence on both of these points in the literature.

1.1 Social Capital and Survey Response

The concept of Social Capital has many incarnations. For example, it can be defined as the degree of reciprocity and trustworthiness within a community (Putnam 1995, 19), the degree of norms and networks that facilitate collective action within a society (Wollcock 2001), or simply a greater presence of goodwill (Hanifan 1916). The link between social capital and survey response probabilities lies in the social integration literature. Specifically, the theory states that the more socially integrated a person is, the more likely he/she is to respond to a survey (Goyder 1987; Groves and Couper 1998).

Activities associated with high levels of social and civic participation (i.e., higher levels of social capital), such as religious, community, political, and volunteer activities, have been linked to positive participation in surveys through the theory of social integration (Woodberry 1998; Groves, Singer, and Corning 2000; Abraham, Maitland, and Bianchi 2006). The more integrated someone feels, the greater the benefit perceived through survey participation (Groves, Cialdini, and Couper 1992; Goyder, Boyer, and Martinelli 2006) and the greater guilt felt through non-participation (as non-participation is perceived as harming the community). In addition, certain types of social capital characteristics, such as trust, may make someone respond more positively to unfamiliar survey takers (Putnam 1995).

In 2000, Census used survey information about voting behavior within an area as a signal

of the degree of social integration of residents in that area. Advertising was targeted to areas with low levels of voting behavior in an effort to compensate for the anticipated lower response rate that would be predicted by lower levels of perceived social integration (GAO 2010, 7). The Census Bureau discovered, however, that this measure of civic participation (voting) was not a strong determinant of survey response.

Amaya and Harring (2017) made use of civic and social engagement reported in the November 2011 supplement to the Current Population survey (CPS) in order to asses the impact of the degree of social integration on the probability of responding to the 2012 American Time Use Survey (ATUS) -- a proportion of those "retiring" from the CPS were approached to also complete the ATUS. CPS respondents with higher civic engagement were more likely to respond to the ATUS, however family and neighborhood engagement did not affect response rates.

The question in this paper is specifically related to mail return rates to the DC, whose baseline is likely quite different from that of other surveys (e.g., the ATUS). In fact, it may be much more difficult to identify systematic relationships between behavior and Census return rates, as was the case in the GAO (2010) report, since the return rate is already very high. Nevertheless, this paper examines the relationship between multiple dimensions of social capital and survey response. In addition to voting behavior, other civic activities are combined to obtain a measure of community involvement, as well as measures of social trust, political involvement, religious activities, and informal social interactions are investigated in order to determine whether other dimensions of social capital are predictive of higher rates of DC mail-in rates.

1.2 Decennial Census Return Rates and Targeted Media

Even if one is able to determine that areas with higher levels of social capital have higher DC return rates, the next obvious question is whether this information is actionable. In other

words, can targeting resources on areas with low social capital improve their return rates? In order to determine the effectiveness of targeting one type of resource on return rates, the Census Bureau performed an experiment during the 2010 decennial census by pairing up demographically and economically similar designated market areas and roughly doubling the exposure of the population in most targeted areas to television and radio media messages related to the DC (Bates, McCue, and Lotti 2012). The experiment was unable to uncover any impact of the additional media exposure on mail-in return rates in the treated areas, relative to the control areas. The authors identify two potential reasons for this. One, the media dosage levels for all areas (treated and control) were already high, meaning that something like the concept of diminishing returns to additional media exposure had likely set in. And, two, the mail-in response rates were already quite high in all areas (treatment and control), so that something stronger than passive media exposure is likely necessary to get the last hold-outs.

The message from this experiment is that even if we can identify some characteristics that would help in the prediction of mail-in census return rates, the type of resources re-assigned will require additional consideration.

2 Methodology

The goal of the analysis is to determine whether knowing something about an area's level of social capital can better predict mail return rates for the 2000 Decennial Census (DC) than not using such information. We undertake two analyses to answer this question. The first is based on the Hard-to-Count (HTC) methodology developed by Census staff in the 1990s (Bruce, Robinson, and Sanders 2000) involves simple tabulation of return rates along multiple area demographics. The second analysis makes use of multivariate regression and follows the

guidelines of a competition sponsored by Census to identify area characteristics that have the greatest power in predicting return rates (Erdman and Bates 2014).

2.1 Data

The analyses contained in this paper make use of three data sources. First, the publicly available Planning Database (PDB) for the 2000 DC contains the dependent variable of interest -- mail return rates by census tract. Some explanatory variables not easily available from the micro level 2000 Decennial Census (DC) database are also obtained from the PDB. Second, the 2000 DC is used to construct additional census tract level characteristics that have been found to be highly correlated with return rates. And, third, since measures of social capital are not available in the DC, we make use of the 2000 Social Capital Community Benchmark Survey (SCCBS) to obtain parameters in order to predict social capital measures for individuals in the DC. The commonly applied technique of using one sample to obtain predictors for an outcome to be applied to a second sample is called two-sample two-stage least squares (2S2SLS) (Ridder and Moffitt 2007), made popular by (Angrist and Krueger 1992, 1995).[3] Details of how measures of social capital are constructed using the SCCBS and applied to DC are found in Appendix A. The net result is that for each census tract, we estimate the probability of that census tract has a high (vs. medium or low) level of each of six social capital measures -- altruism, sociability, religiosity, political activism, political engagement, and trust. These social capital measures are then used to supplement other census tract characteristics found to be important in identifying hard-to-count areas. As seen in Appendix A, prediction of social capital levels is reasonably good for low and high levels of each social capital level (except high levels of political activism).

[3] Also see Inoue and Solon (2010) for the distinction between 2S2SLS and two-sample instrumental variables (2SIV). Other applications of 2S2SLS can be found in Dee and Evans (2003), Carroll, Dynan, and Krane (2003), Currie and Yelowitz (2000), Fang, Keane, and Silverman (2008), and Keane and Stavrunova (2014).

In addition, these predication compare favorably with first-stage estimation results from others employing the 2S2SLS methodology.

Table 1 lists all of the characteristic measures obtained from the DC and the PDB along with their expected correlation with non-return rates, based on Erdman and Bates (2014). Characteristics that are positively correlated with a high non-return rate include median household income; percent of families with children; a young population; percent of the population that is Hispanic, male, and in the labor force; percent of housing units that are renter occupied or vacant; percent below the poverty line; and the average number of people in a household.

[Table 1 about here]

Table 2 lists the mean of each of the social capital measure used for the full sample and for census tracts with below and above median non-return rates. Overall, the average non-return rate across census tracts is about 25 percent -- 17 percent in low non-return rate census tracts and 32 percent in high non-return rate areas. A greater share of census tracts with high levels of all measured social capital characteristics have below-median non-return rates. In other words, there appears to be a positive correlation between high levels of social capital and census return rates. This suggests that knowing something about an area's social capital might help identify hard-to-count areas. Since social capital is likely to be highly correlated with many of the characteristics listed in Table 1, the question is whether social capital contributes any additional information.

2.2 Hard-to-Count (HTC) Scores

The backbone of the efforts by Census to identify areas that are likely to have low return rates, and, hence, deserve greater attention and resources, is called the Hard-to-Count (HTC) score, and was developed by Bruce, Robinson, and Sanders (2000). The basic idea is that each

census tract gets a number of points based on where in the sample distribution it falls for each of the variables considered. Then the point for each characteristic are summed for the census tract's total HTC score. For example, if a census tract is in the 55th percentile for the percent of households in poverty, it will receive a 1 for that variable. If it is in the 95th percentile for percent of single-unit structures, it will receive a 9. The assignment of points is summarized in this table (see Erdman and Bates 2014).

Percentile	50	55	60	65	70	75	80	85	90	95	97	100
Points	0	1	2	3	4	5	6	7	8	9	10	11

If these were the only two characteristics considered, the census tract would have a total score of 10. Since the total HTC score value depends on the number of variables used to construct it, and since we want to compare scores using only the baseline 20 characteristics listed in Table 1 with scores obtained when six additional social capital variables from Table 2 are included, we normalize the score by dividing by the number of variables used to construct it so that we can compare across specifications. Therefore, each of the HTC scores (whether baseline, or including all social capital variables) will have a maximum value of 11.

2.3 Multivariate Regression Analysis

Partially in response to the GAO's call for the Census Bureau to make use of a predictive model to better anticipate those areas with HTC populations (GAO 2010, 32), in 2012 Census launched a world-wide competition to predict mail return rates for the 2010 DC.[4] Contestants made use of publically available data to build their models. Winning entrants made use of complicated machine-learning algorithms and made use of hundreds of predictors, including ones they supplied on their own. Based on this experience, Census decided on two primary criteria that would dictate strategy going forward: the model should be easy to interpret (hence a

[4] The contest is described in more detail in Erdman and Bates (2014).

preference for simple OLS estimation) and the predictors should be actionable (hence the information must be easily observable or collectable).

With these criteria in mind, we conduct a statistical, multivariate regression analysis to consider the predictive power of a model of non-return rates across census tracts that includes social capital variables. The predictive power of this model will be compared with results from a baseline model excluding social capital predictors. Arguably, the model including social capital characteristics does not meet the actionable criterion -- these measures are not gathered in any on-going Census survey. However, depending on the potential gains, the Census may want to include some social capital indicators in the American Community Survey, at least, perhaps, leading up to a new DC.

We model non-response rates at the census tract level.[5] All characteristics and social capital are aggregated to the census tract level. We model the proportion of the households in census tract *t* not returning their form, the census response rate $(1 - R_t)$, as a function of these average characteristics $(\bar{X}_t)$, the average probability that individuals in the census tract have a high (vs. medium or low) level of social capital *j* (see Appendix A for how this is constructed), and interactions of all characteristics with each social capital indicator.

$$(1 - R_t) = \beta_s' \bar{X}_t + \sum_{j=1}^{6} \left(\varphi_j + \gamma_j' \bar{X}_t\right) * \overline{Prob(\widehat{SK}_t^J = high)} + \varepsilon_t \ . \quad (1)$$

There is a possibility that the level of social capital in a census tract is endogenous to the tract's response rate. In other words, there may be omitted variables that affect both the response rate and the level of social capital, such as the crime rate, for example. In addition, a person who has a higher propensity to return the census form may exhibit higher levels of social capital activities. The fact that social capital is not provided in the data set with which we estimate

[5] The dependent variable is the census tract non-return rate, as opposed to the return rate in order to be comparable to the scoring in the HTC model.

census response actually works in our favor here. It is highly unlikely that any of the observations in the SCCBS are the same respondents completing the one percent sample (long-form) of the decennial census, and because the two samples are from the same population, it is akin to applying split sample IV -- we are instrumenting the actual census tract level of social capital with an estimate obtained from a different sample from within the same population.

3 Results

3.1 Hard-to-Count (HTC) Scores

After constructing the HTC scores as described above for each census tract, we then need to decide how to use this information for purposes of devoting resources to reduce that score. Typically, attention is focused on census tracts in the top X% of the HTC score distribution. Table 3 reports the number of census tracts identified in the top 75th, 90th, and 95th percentile of the HTC score distribution. These percentiles serve as potential thresholds above which a census tract would receive additional resources with the goal of lowering their HTC score.

[Table 3 about here]

There are a total of 63,929 census tracts included in this analysis. Using the baseline set of 20 characteristics listed in Table 1, 16,001 (6,543 and 3,278) census tracts find themselves in the top 75th (90th and 95th) percentile of the constructed HTC scores. For comparison, we recalculate the HTC score for each census tract (normalized by the number of characteristics) including all the social capital variables and then each one separately. Mostly, including the social capital variables, there are fewer census tracts (reported in bold) finding themselves in the 90th and 95th percentiles. Religiosity and trust are the two social capital characteristics that consistently flag fewer census tracts for additional resources across the distribution. However,

the gains in identifying fewer census tracts on which to expend resources (i.e., the potential gain in not wasting those resources) is marginal at best. In other words, if it was decided to expend extra resources on census tracts found in the top 90th (95th) percentile of the HTC score, then knowing more about the average social activity (religiosity) in the census tract would identify 71 (80) fewer census tracts on which to expend those resources. This would save some money, but not likely enough to warrant the extra expenditure of gathering information about different levels of social capital by census tract.

3.2 Predicting Census Response Rates through Multivariate Analysis

The full set of parameter estimates from estimating equation (3) are reported in Appendix B, Table B1. The results tell us generally that census tracts with higher non-return rates are those with greater shares of vacant units, households below poverty, black non-Hispanics, and males; with more people per household; and those with higher median incomes and lower education. Lower non-return rates are associated with more single unit structures, married households, older populations, longer-term residents, higher education and labor force participation, and greater shares of Hispanics and non-white Hispanics. These results are mostly consistent with those reported by Erdman and Bates (2014). Differences appear for Hispanics and blacks, although a difference is that they do not make Hispanic an independent category from race. We also differ in the result for less than high school education for uncertain reasons, other than the fact that we include different age regressors than Erdman and Bates.

Table 4 presents the results from the multivariate regression that correspond to the social capital variables. The first column is from the specification that includes all social capital variables, but no interactions with other census tract characteristics. The second specification includes all social capital variables as well as their interactions with all of the other (non-social

capital) characteristics; hence the reporting of the marginal effect, which takes the interaction terms into account. Focusing on the most exhaustive specification (the one including interactions), we find that higher levels of religiosity, trust, and political engagement are associated with lower census tract non-response rates. This is consistent with findings of Woodberry (1998); Groves, Cialdini, and Couper (1992); and Abraham, Maitland, and Bianchi (2006). We also find higher non-response rates in census tracts with higher levels of political activism, which could make sense if that activism is fueled by distrust of government. Our finding that lower non-response rates in census tracts with high community involvement in the model without interactions is consistent with Amaya and Harring (2017). Since that result goes to zero when interacted with all other census tract characteristics suggests that civic involvement is particularly highly correlated with those other characteristics. In addition, whereas Amaya and Harring (2017) did not find any relationship between family and neighborhood involvement (sociability), we found that higher levels of that social capital is associated with *increased* non-response.

[Table 4 about here]

Again, the challenge is how to use these estimation results to inform the question about whether it would be worth adding information about social capital characteristics to improve the identification of hard-to-count census tract. Toward that end, Table 5 reports Type I and Type II errors in predicting census tracts in the top 90 and 95 percent of the non-reporting distribution. The first column reports the adjusted R squared from each OLS regression, which is essentially the criterion used by the Census competition. Un-surprisingly, the model that explains most of the variation in non-return rates across census tracts is the one that includes all social capital variables plus all the social capital variables interacted with all of the baseline census tract

characteristics. One implication of this is that the social capital variables are not perfectly co-linear with the baseline characteristics. The next highest adjusted R squared statistic is found from the regression that includes only the trust social capital variables and its interaction. All of the regressions that include social capital characteristics, in one form or another, produce a better fit of the data than the 20 baseline characteristic regressors by themselves.

[Table 5 about here]

The specification including all social capital variables and their interactions also reduces both Type I and Type II errors in predicting census tracts falling in the top 90 and 95 percent of the non-return rate distribution. The two social capital variables that reduces these prediction errors by the greatest amount on their own (including interaction) are trust and social activity. But, again, the marginal gains are questionable. Focusing on trust, since that social capital variable also yielded the best fit (among individual social capital variables), knowing something about the average trust levels of residents in census tracts reduces the Type I and Type II errors of census tracts in the top 90th (95th) percentile both by seven (five) percent, compared with predictions not using that information (i.e., just using the 20 baseline characteristics).

4 Conclusions and Implications

Two analyses in this paper find that non-return rates for the Decennial Census are lower in census tracts with higher levels of certain social capital characteristics. When social capital characteristics are included in the construction of census tract Hard-to-Count (HTC) statistics, however, at best the modified HTC flags 90 fewer census tracts at the 90th HTC percentile, and 80 fewer census tracts at the 95th percentile. It's unclear whether the additional resources needed

to identify those social capital characteristics is less than the resources saved by targeting fewer census tracts for additional resources designed to improve return rates.

One comes to essentially the same conclusion assessing the results from the multivariate analysis. However, the multivariate analysis does allow one to choose their direction of tolerance -- is it worse to identify a census tract as HTC when it shouldn't be (Type II error), or is it worse to miss a HTC census tract when it would benefit from additional promotional resources (Type I error)? Focusing on the 90th percentile, the baseline model has a Type I error of 38.75 percent -- 38.75 percent of all census tracts are in the top 90th percentile of non-return rates, but are predicted below the 90th percentile. In addition, the baseline models suffers from a 4.23 percent Type II error -- 4.23 percent of all census tracts are predicted to be in the 90th percentile when they are actually below the 90th percentile. For all models the Type I error is larger than the Type II error. As was mentioned above, including a measure of a census tract's average trust level (along with its interactions) yields the highest adjusted R squared test statistic. The Type I error (for the 90th percentile), however, is only reduced by two percentage points and the Type II error is reduced by 0.23 percentage points (both amount to about a five percent reduction). Again, this begs the question of whether this small reduction in error is worth the resources that would be required to obtain measures of social capital. This question is especially relevant given Census' own finding that additional media exposure in targeted areas did not impact mail-in return rates (Bates, McCue, and Lotti 2012).

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Table 1. Variables used in the analyses, their source, and expected correlation with non-return rates at the census tract level.

Variables	Expected Correlation with Non-return Rate
2000 Decennial Census (DC)	
Median household income	+
Percent living in the same house at least five years	-
Percent of population 25 years and older with less than HS education	-
Percent of population 25 years and older with college plus education	-
Percent of families with own children	+
Percent of population ages 5-17	+
Percent of population ages 65 plus	-
Median age	-
Percent married households	-
Percent of population that is White, non-Hispanic	-
Percent of population that is Black, non-Hispanic	-
Percent of population that is Hispanic	+
Percent of population that is male	+
Population density	-
Percent of population 16 years and older in the labor force	+
2000 Planning Data Base (PDB)	
Nonreturn rate	na
Percent of units that are renter occupied	+
Percent of housing units that are vacant	+
Percent of single unit structures	-
Percent of households below the poverty level	+
Average number of persons in households	+

Notes: Expected correlation derived from Erdman and Bates (2014, Table 2).

Table 2. Selected sample means of census tract average non-return rates and social capital variables by high levels of non-return rates.

Variable	Full Sample	Non-return Rate < Median	Non-return Rate ≥ Median
Average non-return rate across CTs	24.65	16.98	32.32
	(9.845)	(3.945)	(7.787)
Prob Religiosity is high	34.72	34.96	34.47
	(7.921)	(6.880)	(8.834)
Prob Trust is high	29.56	36.64	22.48
	(11.87)	(8.693)	(10.28)
Prob Political Activism is high	18.00	19.24	16.76
	(6.987)	(6.436)	(7.290)
Prob Political Engagement is high	29.17	34.66	23.69
	(10.02)	(8.488)	(8.295)
Prob Community Involvement is high	28.88	31.16	26.60
	(7.763)	(7.834)	(6.984)
Prob Sociability is high	31.17	32.78	29.57
	(5.993)	(4.660)	(6.707)
Observations	63,929	31,962	31,967

Notes: Standard deviations in parentheses. Note that, "These mail return rates for tracts are unofficial: they are not consistent with published Census 2000 response rates for tracts. The tract-level rates used for the Planning Database are based on tabulation geography, while the published rates are based on collection geography," (Bruce and Robinson undated, 13).

Table 3. Number of census tracts with HTC scores in the 75th, 90th, and 95th percentiles for each specification.

Regression Model	Number in the 75th HTC score percentile	Number in the 90th HTC score percentile	Number in the 95th HTC score percentile
Baseline (20 variables listed in Table 1 without any SK variables)	*16,001*	*6,542*	*3,278*
All SK variables	16,111	**6,452**	3,299
Religiosity only	**15,997**	**6,515**	**3,198**
Trust only	**15,984**	**6,527**	**3,276**
Political Activism only	16,300	6,563	**3,255**
Political engagement only	16,252	**6,476**	3,316
Community activity only	16,247	**6,527**	**3,246**
Social activity only	16,057	**6,471**	**3,268**

Notes: Total number of census tracts used in estimations is 63,929. The Mail Non-Return Rate is calculated by dividing the number of forms mailed to occupied units that were not returned by the number of occupied units in the mailout universe.

Table 4. OLS parameter estimates and marginal effects for predicting a census tract's average non-return rate.

VARIABLES	Model including all Social Capital Regressors, no interactions; parameter estimate	Model including all Social Capital Regressors, with interactions; marginal effect
Prob Religiosity is high	0.0026	-0.035***
	(0.0053)	(0.006)
Prob Trust is high	-0.2154***	-0.125***
	(0.0073)	(0.009)
Prob Political Activism is high	0.1334***	0.162***
	(0.0083)	(0.009)
Prob Political Engagement is high	0.0097	-0.052***
	(0.0089)	(0.010)
Prob Community Involvement is high	-0.0419***	0.003
	(0.0091)	(0.010)
Prob Sociability is high	0.1956***	0.110***
	(0.0119)	(0.013)
Observations	63,929	63,929
R-squared	0.6845	0.7373
MSE	30.58	25.46

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Dependent variable is shares of forms in CT occupied housing not returned by mail. See Table B1 in Appendix B for the full set of regression results, including other census tract characteristics.

Table 5. Incidence of Type I and Type II error in predicting census tracts falling into the top 90% and 95% of the non-return rate distribution.

Regression Model	Adjusted R squared from the OLS regression	Type I Error; % of CT actually in 95th percentile predicted not to be there	Type II Error; % of CT not in 95th percentile predicted to be there	Type I Error; % of CT actually in 90th percentile predicted not to be there	Type II Error; % of CT not in 90th percentile predicted to be there
Baseline (20 variables listed above without any SK variables)	*0.6725*	*50.25%*	*2.64%*	*38.75%*	*4.23%*
All SK variables	0.6845	49.47	2.60	38.16	4.16
All SK variables and their interactions with the other 20 regressors	**0.7373**	**41.36**	**2.17**	**33.31**	**3.61**
Religiosity only	0.6727	50.25	2.64	38.73	4.22
Trust only	0.6762	50.31	2.64	38.33	4.18
Political Activism only	0.6754	50.09	2.63	38.90	4.24
Political engagement only	0.6726	50.19	2.63	38.65	4.22
Community activity only	0.6749	50.22	2.64	38.70	4.22
Social activity only	0.6743	49.66	2.61	38.36	4.18
Religiosity and interactions	0.6797	49.53	2.60	38.13	4.16
Trust and interactions	**0.6999**	**46.76**	**2.45**	36.74	4.00
Political Activism and interactions	0.6901	49.10	2.58	37.86	4.13
Political engagement and interactions	0.6947	47.66	2.50	37.02	4.03
Community activity and interactions	0.6890	48.66	2.55	37.53	4.09
Social activity and interactions	0.6937	48.00	2.52	**36.06**	**3.92**

Notes: Total number of census tracts used in estimations is 63,929. The Mail Non-return Rate is calculated by dividing the number of forms mailed to occupied units that were not returned by the number of occupied units in the mailout universe.

Appendix A: Using Two-Sample Two-Stage Least Squares to Estimate Census Tract Level Social Capital

In 2000 the Roper foundation conducted a national survey, the Social Capital Community Benchmark Survey (SCCBS), to gauge the level of a multitude of dimensions of individual social capital.[6] We use the SCCBS as the source for estimating social capital determinants at the individual level. The level of social capital in a census tract is determined by the social capital level of its inhabitants, so that is where we start. Parameter estimates are obtained from the SCCBS data that can be used to predict social capital for individuals in the DC. Fortunately, the SCCBS and the DC are fairly harmonious with respect to their measures of demographics. This is fortunate, since we are restricted to those variables that are found in both surveys in order to use the estimated parameters from one data set to predict social capital in the second.

A.1 Creating Weights to use in the SCCBS. Since we are predicting out of sample, and in spite of the fact that both the DC and the SCCBS are both national surveys, we are also interested in how the two samples compare in their distributions across demographics. In other words, we want to be sure that the parameter estimates obtained from the SCCBS sample are likely to be applicable, at least at the means, to observations in the DC. Both the SCCBS and DC surveys contain individual weights designed to generate a random national sample. Using the survey-provided sample weights, 96 percent of the weighted means of the common variables in the DC and SCCBS were statistically different from one another at least at the 95 percent confidence level.

To estimate the social capital equation on a sample that is more representative of the DC (for which the prediction will be made), we use an inverse probability weighting methodology,

[6] There is a more recent 2006 SCCBS survey, however there is no confidential version and no indicator for census tract, which is important for constructing an instrument for individual social capital.(see https://ropercenter.cornell.edu/2006-social-capital-community-benchmark-survey/).

akin to the one used in DiNardo, Fortin, and Lemieux (1996), in order to create a counterfactual distribution of the SCCBS that is much more similar to the DC. This amounts to estimating, in the combined DC and SCCBS samples, the probability of an observation being observed in SCCBS, using as explanatory variables as many demographics and their cross-multiples as is feasible:

$$P(observation\ i\ \in SCCBS|X) = \Lambda(X'b)\ . \qquad \text{(A1)}$$

The parameter estimates from this logit model are then used to construct the inverse probability ratio, $\frac{\Lambda(X'\hat{b})}{1-\Lambda(X'\hat{b})}$, for each observation in the SCCBS. This is the re-weighting function used to modify the individual weight provided in the SCCBS.

With over 20 million observations in the DC, even small practical differences in means will be statistically significant. However, there is dramatic improvement using the inverse-weighted adjustment to the means in the SCCBS. Using the new weights, the percent of common variables that are statistically different from one another is reduced to 64 percent -- with 85 percent of the re-weighted means of all variables being significantly closer to the DC mean than they were using the survey supplied weights.[7]

A.2 Identifying a Person's Unobserved Social Capital Through Factor Analysis. A person's social capital is not a characteristic that is observed. In addition, there are many dimensions to social capital, from sociability, altruism, political engagement, etc. (for example, see Putnam 1995). The SCCBS contains a multitude of questions designed to elicit, based on observed/reported activity, the level of these different dimensions of social capital. For the analysis in this paper, we identify five dimensions of social capital in order to be as inclusive of as many behaviors as possible. We perform a factor analysis using the responses to a variety of

[7] More details of the means comparisons are suppressed for confidentiality reasons.

questions in the SCCBS in order to identify each person's latent degree of social capital across the dimensions of religiosity, trust, political activism, political engagement, altruism, and sociability. Factor analysis is a data reduction technique for expressing observed variables as linear combinations of a few unobserved variables called *factors*.

We conduct factor analysis within each of the categories and retain the first factor extracted from each analysis. It is typical for factor analysis to produce as many factors as the number of original variables in a particular category. As such, the determination of how many factors to retain from the factor analysis for subsequent analyses is based on well-known rules of thumb (e.g., see Rencher 1997). One is to retain all factors with eigenvalues greater than one. Another is to choose the number of factors required to reach a given percentage of explained total variation (measured as trace of correlation matrix). Other researchers use statistical hypothesis testing when the maximum likelihood method of estimation of the loadings matrix is adopted, while others simply retain the first factor extracted (e.g., see Deller et al. 2001). We followed the last rule of thumb in this paper, since a driving motivation for performing factor analysis in this case was for the purposes of data reduction.

For each dimension of social capital, we perform factor analysis using the principal component method on the polychoric correlation matrix since the variables in our data are either binary, ordinal, or continuous. Table A1-A6 summarize results from our factor analysis. Religiosity has the highest cumulative variance explained by the retained factor (0.76), whereas sociability has the lowest (0.31).

[Table A1 - A6 about here]

A.3 Estimating Social Capital. The social capital factors identified are continuous individual variables. Since we have little hope of adequately fitting those continuous variables,

and since the metric of each measure is uninterpretable anyway, we create low, medium, and high categories of each factor by splitting the distribution of the sample into thirds. Since the categories created are ordered from lower to higher levels of the social capital variable in question, we then estimate the parameters of each social capital equation as an ordered logit.[8] The probability that individual *i*, living in census tract *t*, has social capital level *k* of type *j* (*j*=religiosity, trust, political activism, political engagement, community involvement, and sociability) is formally expressed as:

$$Pr[SK_{it}^{j} = k] = Pr[\mu_{k-1} < \alpha_0 + \mathrm{A}'X_i + \mathrm{B}'Y_i + \mathrm{C}'Z_t + u_{it} \leq \mu_k] \,, \qquad \text{(A2)}$$

where u_{it} is assumed to be logistically distributed and the estimated cutpoints μ=1,2 separate three possible outcomes *k*= low, medium, high for each type of social capital.

Regressors X_i reflect person *i*'s demographics. Regressors Y_i are additional determinants of a person's level of social capital. Regressors unique to the social capital equation are reflected through Z_t. For these identifying regressors, we construct average census tract characteristics for each observation. The theory is that the characteristics of those in close proximity influence one's level of social capital. There is some concern that demographics *in* a person's census tract may be endogenous to that person's level of social and community involvement. In other words, there are unobservable factors both affecting a person's level of social capital and the type of people residing in the location where that person has chosen to live (e.g., someone with high levels of social capital may choose to locate in that census tract because of the high degree of activity by social and community/civic organizations). This potential for endogeneity is why characteristics in *surrounding* census tracts will be used as identifying regressors, rather than the value of those variables *in* the person's own census tract. These surrounding characteristics will be weighted by

[8] We also estimated a multinomial logit, but the results and fit did not change.

the distance (from centroids) of the census tract from that person's census tract. Census tracts in the person's own and surrounding states will be used to construct the average. This method of construction of an instrument in the face of potential geographic endogeneity has been widely applied in the empirical literature (for example, see Lee and Gordon 2005).

Once the level of social capital is predicted for each person in the DC, using the parameter estimates from the ordered logit estimation, an average probability for low, medium, and high levels of each measure of social capital is calculated at the census tract level. Then the average probability that a census tract has a high level of each social capital measure (as opposed to medium or low) will be used with other characteristics to identify hard-to-count census tracts.

A.4 Social Capital Estimation Results

Table A7 contains the estimation results from the social capital ordered logit analysis. The results link both individual characteristics and the distance weighted census-tract "neighborhood" characteristics with each social capital outcome. Using the same data set, Brueckner and Largey (2008) estimate two types of social capital -- friendship oriented variables (closest to our sociability) and group-involvement variables (closest to our community involvement). Our results are fairly consistent with their findings.

[Table A7 about here]

Overall, married individuals have higher levels of all social capital, except sociability. Higher income, higher education, and being a citizen are related to higher levels of social capital (also see DiPasquale and Glaeser 1999; Helliwell and Putnam 1999; Glaeser, Laibson, and Sacerdote 2002). Having lived in an area five years or less and being unemployed are associated with lower levels of social capital (also see Schiff 1992; Glaeser et al. 2000; Putnam 1995). Other characteristics have varying relationships with social capital of different types. Also note

that the identifying regressors as a group are statistically significantly different from zero in each social capital equation and there is at least one regressor individually significant.

Since the parameter estimates from equation (A2) will be used to predict levels of social capital out of sample for individuals in the DC, it's important that we have some confidence in the predictive power of the regressors. Table A8 reports, for each actual (low, medium, high) category the person is in for each social capital measure, the percent of the sample predicted to be in that category. There is clear dominance in correctly predicting the low and high categories (except for political activism high category) of all measures of social capital. However, individuals with actual medium values of each social capital are fairly equally likely to be predicted in the low and high categories as they are to be predicted in the medium category.

[Table A8 about here]

Table A8 also reports the correlation between the actual and predicted social capital categories within the SCCBS sample.[9] The correlations range from 0.28 for political activism to 0.56 for political engagement; these are quite reasonable compared to the reported fist-stage correlation coefficients obtained by others' applications in an OLS framework of this two-sample prediction strategy. Dee and Evans (2003) make use of predictions from a first stage estimation with r-squared statistics less than 0.02; Carroll, Dynan, and Krane (2003) report first-stage adjusted r-squared test statistics between 0.28 and 0.48; Currie and Yelowitz (2000) report a first-stage r-squared statistic less than 0.1; Fang, Keane, and Silverman (2008) and Keane and Stavrunova (2014) contain similar analyses using the same data and fist-stage r-squared statistics ranging from 0.02 to 0.25. The focus of these authors on their first-stage estimations is the degree to which the excluded regressors (those not included in the second-stage regression) were

[9] Since the first stage estimation is an ordered logit, we don't get the usual fit diagnostic of an r-squared test statistic we would get if the first stage were estimated via ordinary least squares.

statistically significantly different from zero (also see Carroll, Dynan, and Krane 2003); again, Chi squared tests indicate that our excluded regressors, the group of distance weighted characteristics of surrounding census tracts, is statistically significantly different from zero.

Table A5. Factor analysis results for religiosity social capital factor.

Variable from SCCBS	Description	Factor Loadings	Correlation between retained factor and original variables	Uniqueness
RELMEM2	Church or synagogue member	0.8987	0.8097	0.1923
RELATEN2	How often attend religious service	0.9166	0.8641	0.1598
RELPART2	Participate in church atctivities besides attending service	0.9178	0.8199	0.1577
GRPREL	Participate in organization affiliated with religion	0.7348	0.5808	0.4601

Notes: Cumulative Variance Explained by retained factor = 0.7575. Eigenvalue of retained factor = 3.0313.

Table A2. Factor analysis results for trust social capital factor.

Variable from SCCBS	Description	Factor Loadings	Correlation between retained factor and original variables	Uniqueness
TRUST2	Whether most people can be trusted	0.7453	0.6592	0.4445
TR2NEI	How much trust neighbors	0.7827	0.7345	0.3874
TR2SHOP	How much trust people in stores you shop	0.7797	0.7416	0.3920
RACETRST	How much trust people of other race	0.7168	0.6697	0.4861
TR2NEWS	How much trust local news media	0.7751	0.7619	0.3992
TR2COP	How much trust local community police	0.5541	0.5280	0.6930
CONFIDE	Number of people you can confide in	0.3848	0.3398	0.8519

Notes: Cumulative Variance Explained by retained factor = 0.4780. Eigenvalue of retained factor = 3.3459.

Table A3. Factor analysis results for political activism social capital factor.

Variable from SCCBS	Description	Factor Loadings	Correlation between retained factor and original variables	Uniqueness
PETITION	Signed a petition in past 12 months	0.6399	0.5156	0.5905
RALLY	Attended a political rally or meeting in past 12 months	0.7816	0.6530	0.3890
MARCH	Participated in demonstrations, boycotts, marches	0.7609	0.5922	0.4211
GRPPOL	Participate in political group	0.8194	0.6680	0.3286
REFORM	Belong to any group that took local action for reform	0.7984	0.6674	0.3625
GRPETH	Participate in ethnic, nationality, or civil rights organization	0.7305	0.5656	0.4664

Notes: Cumulative Variance Explained by retained factor = 0.5736. Eigenvalue of retained factor = 3.4419.

Table A4. Factor analysis results for political engagement social capital factor.

Variable from SCCBS	Description	Factor Loadings	Correlation between retained factor and original variables	Uniqueness
PAPER	Days in past week respondent read a daily newspaper	0.5743	0.5136	0.6702
REGVOTE	Currently registered to vote	0.9350	0.8507	0.1257
VOTEUS	Voted in 1996 presidential election	0.9355	0.8605	0.1248

Notes: Cumulative Variance Explained by retained factor = 0.6931. Eigenvalue of retained factor = 2.0793.

Table A5. Factor analysis results for community involvement social capital factor.

Variable from SCCBS	Description	Factor Loadings	Correlation between retained factor and original variables	Uniqueness
GRPPTA	Participate in parent association or other school support	0.5633	0.4919	0.6826
GRPNEI	Participate in neighborhood association	0.5631	0.4830	0.6830
GRPSOC	Participate in social or welfare organization	0.7201	0.6334	0.4815
GRPFRAT	Participate in service or fraternal organization	0.5526	0.4541	0.6947
PUBMEET2	How often attended a public meeting discussing school	0.5133	0.5061	0.7366
VOLTIME2	Number of times volunteered	0.5740	0.5750	0.6705
PROJECT	Worked on community project in past 12 months	0.8221	0.7191	0.3241

Notes: Cumulative Variance Explained by retained factor = 0.3896. Eigenvalue of retained factor = 2.7271.

Table A6. Factor analysis results for sociability social capital factor.

Variable from SCCBS	Description	Factor Loadings	Correlation between retained factor and original variables	Uniqueness
GRPSPORT	Participate in sports club, league, or outdoor activity	0.3990	0.3525	0.8408
FRNDHOM2	How often had friends over to your home	0.7067	0.7069	0.5005
FRNDHNG2	How often hung out with friends in a public place	0.6895	0.6888	0.5246
PARADE2	How often attended parade, local sports or arts event	0.5657	0.5640	0.6800
CARDS2	How often played cards or board games with others	0.5553	0.5565	0.6916
FRIENDS	Number of close friends	0.4468	0.4302	0.8004
ARTIST2	How often took part in artistic activity with group	0.4571	0.4576	0.7911

Notes: Cumulative Variance Explained by retained factor = 0.3101. Eigenvalue of retained factor = 2.1709.

Table A7. Ordered logit social capital estimation results.

VARIABLES	Religiosity	Trust	Political Activism	Political Engagement	Community Involvement	Sociability
Age	-0.0506	0.0745	-0.0398	-0.0378	-0.0814	-0.2913***
	(0.0682)	(0.0753)	(0.0785)	(0.0869)	(0.0712)	(0.0629)
Age squared	0.0014	-0.0011	0.0014	0.0020	0.0026*	0.0050***
	(0.0015)	(0.0016)	(0.0017)	(0.0019)	(0.0015)	(0.0014)
Married=0,1	0.5124***	0.1571***	0.0651	0.3556***	0.1035*	-0.1566***
	(0.0567)	(0.0609)	(0.0580)	(0.0643)	(0.0577)	(0.0547)
Number of children in HH	0.1686***	-0.0412	0.0333	-0.0387*	0.2685***	0.0229
	(0.0228)	(0.0270)	(0.0256)	(0.0224)	(0.0227)	(0.0210)
Household total income GE $30,000=0,1	0.1113*	0.3571***	0.2697***	0.2144***	0.4446***	0.3713***
	(0.0646)	(0.0724)	(0.0698)	(0.0730)	(0.0636)	(0.0638)
High school education=0,1	0.3427***	0.1574	0.4128***	1.128***	0.3701***	0.1713
	(0.1160)	(0.1306)	(0.1379)	(0.1423)	(0.1087)	(0.1046)
Some college education=0,1	0.7142***	0.3850***	0.9480***	1.590***	0.8893***	0.4787***
	(0.1190)	(0.1328)	(0.1387)	(0.1364)	(0.1107)	(0.1064)
College graduate=0,1	0.8269***	0.9116***	1.694***	2.187***	1.507***	0.7959***
	(0.1487)	(0.1747)	(0.1709)	(0.1702)	(0.1491)	(0.1479)
Hispanic=0,1	0.5745***	-0.5615***	-0.1657	-0.1134	-0.1922	-0.5562***
	(0.1191)	(0.1466)	(0.1479)	(0.1491)	(0.1204)	(0.1299)
Black, non-Hispanic=0,1	0.9174***	-0.8511***	0.1743	0.3238**	0.2456**	-0.7620***
	(0.1076)	(0.1330)	(0.1239)	(0.1358)	(0.1107)	(0.1197)
Other race, non-Hispanic=0,1	0.1671	-0.1393	-0.4601**	-0.5499**	-0.2978	-0.8797***
	(0.1899)	(0.2346)	(0.2137)	(0.2479)	(0.1944)	(0.2108)
Unemployed=0,1	-0.3400***	-0.2584**	-0.2326*	-0.2929*	-0.3263***	-0.2607**
	(0.1041)	(0.1122)	(0.1203)	(0.1588)	(0.1171)	(0.1096)
Not in the labor force=0,1	0.1067	0.0464	-0.0238	0.0544	0.0231	0.1850***
	(0.0675)	(0.0770)	(0.0701)	(0.0737)	(0.0677)	(0.0642)
Citizen=0,1	0.1800	0.4270***	0.8859***	4.025***	0.6988***	0.5086***
	(0.1150)	(0.1633)	(0.2066)	(0.3153)	(0.1305)	(0.1338)
Lived in area 5 years or less=0,1	-0.2309***	-0.1133**	-0.2772***	-0.2698***	-0.2686***	-0.2746***

VARIABLES	Religiosity	Trust	Political Activism	Political Engagement	Community Involvement	Sociability
	(0.0489)	(0.0534)	(0.0525)	(0.0535)	(0.0531)	(0.0518)
Own home=0,1	-0.2204**	0.3175**	0.2516*	0.3692***	0.2570**	0.1165
	(0.0977)	(0.1276)	(0.1291)	(0.1194)	(0.1053)	(0.1142)
Female=0,1	0.2498***	0.2348***	-0.0649	-0.1720***	0.2358***	0.0085
	(0.0501)	(0.0526)	(0.0491)	(0.0531)	(0.0477)	(0.0466)
Live in MSA=0,1	0.1629*	-0.1278	0.0842	0.0750	0.0396	-0.1543*
	(0.0951)	(0.0955)	(0.0954)	(0.1000)	(0.0882)	(0.0880)
Mid Atlantic region=0,1	0.2312	-0.0978	-0.5144***	-0.2317	-0.1479	-0.0118
	(0.1726)	(0.1749)	(0.1753)	(0.1810)	(0.1696)	(0.1637)
East North Central region=0,1	0.2120	0.1869	-0.2008	0.0187	0.1989	0.1446
	(0.1924)	(0.2026)	(0.1958)	(0.2206)	(0.1859)	(0.1827)
West North Central region=0,1	0.0433	-0.1748	-0.3379	-0.2597	0.1105	0.0188
	(0.2684)	(0.2764)	(0.2680)	(0.2838)	(0.2563)	(0.2642)
South Atlantic region=0,1	0.3343	-0.0918	-0.2242	-0.0177	0.3475	-0.1050
	(0.2127)	(0.2454)	(0.2295)	(0.2303)	(0.2231)	(0.2067)
East South Central region=0,1	0.4734*	-0.0512	-0.1285	-0.0771	0.2488	0.0074
	(0.2576)	(0.2810)	(0.2792)	(0.2695)	(0.2753)	(0.2533)
West South Central region=0,1	0.1512	-0.0090	-0.3255	0.2039	0.4857*	0.4575*
	(0.2585)	(0.2933)	(0.2818)	(0.2512)	(0.2747)	(0.2515)
Mountain region=0,1	-0.2195	0.3206	0.4199*	0.2713	0.4145*	0.1891
	(0.2499)	(0.2765)	(0.2470)	(0.2413)	(0.2512)	(0.2371)
Pacific region=0,1	-0.0759	0.1666	0.4486*	0.4159*	0.1089	0.0883
	(0.2288)	(0.2597)	(0.2348)	(0.2264)	(0.2418)	(0.2189)
College grad * white non-Hispanic	0.0676	0.3372**	-0.0390	0.1527	0.0195	-0.0047
	(0.1239)	(0.1339)	(0.1268)	(0.1259)	(0.1173)	(0.1212)
College grad * own home	0.1216	-0.3619***	-0.2625**	-0.2063*	0.0026	-0.2519**
	(0.1051)	(0.1056)	(0.1030)	(0.1055)	(0.0974)	(0.0994)
White non-Hispanic * own home	0.3994***	0.0342	-0.0852	0.1787	0.0449	-0.2433**
	(0.1166)	(0.1410)	(0.1391)	(0.1408)	(0.1172)	(0.1227)
Age GE 75 years	-0.1120	-0.2468	0.0673	-0.0921	0.3740	-0.2159
	(0.2701)	(0.3309)	(0.3365)	(0.3059)	(0.2954)	(0.2749)

VARIABLES	Religiosity	Trust	Political Activism	Political Engagement	Community Involvement	Sociability
Age LT 25 years	0.1467	0.1439	0.2382	-0.6629***	0.2227	0.0400
	(0.1429)	(0.1565)	(0.1683)	(0.1741)	(0.1426)	(0.1462)
Age cubed (0000)	-0.0983	0.0804	-0.1312	-0.1375	-0.2150**	-0.2874***
	(0.1029)	(0.1129)	(0.1195)	(0.1349)	(0.1070)	(0.0937)
Census Tract Weighted Characteristics						
Share of workers in broad SK occupations	-2.698	12.34**	17.17***	5.981	9.937*	6.433
	(5.600)	(6.253)	(5.909)	(6.027)	(5.513)	(5.2567)
Share of workers in SK industries	-3.366	2.679	55.07***	-8.757	42.46***	22.21
	(14.78)	(15.88)	(15.87)	(18.17)	(16.24)	(15.85)
Labor force participation rate	-4.613	9.591	17.66*	-6.281	2.871	-6.141
	(8.786)	(8.829)	(9.043)	(10.14)	(9.007)	(8.345)
Unemployment rate	-45.46*	-65.12**	16.23	-19.33	19.60	-12.81
	(27.02)	(29.16)	(25.38)	(29.86)	(25.85)	(24.26)
Percent lived in house at least 5 years	0.9137	0.3399	-2.333	2.876	-4.868***	-4.665***
	(1.647)	(1.978)	(1.710)	(2.025)	(1.770)	(1.671)
Median age	-0.0622	0.1037*	-0.1296**	0.0201	0.0571	0.0746
	(0.0518)	(0.0532)	(0.0533)	(0.0576)	(0.0533)	(0.0509)
Diversity index	2.870***	0.3089	-1.208	0.0890	0.7660	0.0611
	(0.9207)	(1.035)	(0.9587)	(0.8824)	(0.9392)	(0.9216)
Female labor force participation rate	5.613	-7.326	-13.089	9.872	-1.212	5.755
	(8.015)	(7.904)	(8.108)	(9.109)	(8.150)	(7.720)
Percent college graduates, 25 and older	-2.338	-0.1786	-5.262**	4.945*	-3.744	-3.1386
	(2.431)	(2.872)	(2.480)	(2.712)	(2.428)	(2.230)
Percent married households	-0.7772	-0.4433	-2.006	-1.325	-1.873	-1.519
	(2.061)	(2.380)	(2.094)	(2.280)	(2.161)	(2.109)
Percent of families with children	1.681	3.979	-11.31***	0.7911	-0.8185	-0.5977
	(3.058)	(3.401)	(3.206)	(3.384)	(2.976)	(2.919)
Percent who own home	4.015***	1.575	-0.9304	0.3394	0.2999	1.103
	(1.556)	(1.698)	(1.703)	(1.714)	(1.664)	(1.617)
Median household income ($00000)	-1.621	1.271	2.568	-3.800*	2.099	2.917*
	(1.687)	(2.135)	(1.780)	(1.961)	(1.782)	(1.658)

VARIABLES	Religiosity	Trust	Political Activism	Political Engagement	Community Involvement	Sociability
Population density (1000/sq mi)	0.0817**	0.0080	0.0223	0.0200	0.0136	0.0352
	(0.0382)	(0.0437)	(0.0465)	(0.0505)	(0.0473)	(0.0437)
Share of population living in MSA	-0.8191*	-0.3714	-0.3052	0.9244*	-0.7635*	-0.7051*
	(0.4591)	(0.4403)	(0.4230)	(0.4879)	(0.4098)	(0.4025)
Constant cut1	-0.2435	11.33***	-4.714	9.702**	2.871	-4.948
	(3.479)	(4.016)	(3.614)	(4.265)	(3.563)	(3.454)
Constant cut2	1.320	12.99***	-3.114	11.74***	4.576	-3.416
	(3.480)	(4.019)	(3.616)	(4.268)	(3.565)	(3.454)
Observations	18,000	18,000	18,000	18,000	18,000	18,000
Chi Sq, H0: CT weighted variables = 0	82.28	111.4	88.94	57.65	32.98	22.67

Notes: Data used for analysis are those from the Social Capital Benchmark Survey (external data file). Dependent variable = 0,1,2. Observations are weighted using an inverse probability adjustment to the weights supplied by the SCCBS (see DiNardo, Fortin, and Lemieux 1996). Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Sample sizes are rounded to the nearest 1,000 for disclosure purposes.

Table A8. Actual versus predicted categories of social capital measures.

VARIABLES	Religiosity	Trust	Political Activism	Political Engagement	Community Involvement	Sociability
Actual Level of Low						
Predicted level of Low	45.21%	60.36%	85.22%	61.56%	54.76%	59.96%
Predicted level of Medium	25.87%	24.13%	12.46%	30.92%	32.69%	24.02%
Predicted level of High	28.91%	15.51%	2.32%	7.52%	12.55%	16.02%
Actual Level of Medium						
Predicted level of Low	29.72%	29.52%	67.93%	16.21%	32.38%	42.01%
Predicted level of Medium	26.26%	33.18%	24.56%	52.92%	40.67%	30.57%
Predicted level of High	44.02%	37.30%	7.51%	30.88%	26.94%	27.42%
Actual Level of High						
Predicted level of Low	12.24%	14.25%	55.63%	6.86%	15.55%	26.60%
Predicted level of Medium	21.53%	31.51%	30.52%	33.61%	34.29%	29.10%
Predicted level of High	64.24%	54.25%	13.85%	59.53%	50.16%	44.30%
Correlations between actual and predicted categories across individuals	0.32	0.41	0.28	0.56	0.39	0.30

Appendix B: Full set of Parameter Estimates for Multivariate Analysis.

Table B1. OLS estimates and marginal effects for predicting a census tract's average non-return rate.

VARIABLES	No Social Capital Regressors (1)	All Social Capital Regressors, no interactions (2)	All Social Capital Regressors, with interactions (3)
Renter occupied units	0.0117***	0.0051	-0.0442
	(0.0045)	(0.0044)	(0.0402)
Vacant units	0.1537***	0.1632***	0.2720***
	(0.0057)	(0.0057)	(0.0618)
Single unit structures	-0.1215***	-0.1156***	-0.1153***
	(0.0026)	(0.0025)	(0.0243)
Percent below poverty	0.0941***	0.0998***	0.3367***
	(0.0083)	(0.0084)	(0.0630)
Percent married households	-0.0282***	-0.0091	-0.2034***
	(0.0062)	(0.0067)	(0.0491)
Average number of persons per household	0.6962**	0.4638*	-5.857**
	(0.2733)	(0.2745)	(2.371)
Median HH income ($0000)	0.2697***	0.1693***	-0.7269**
	(0.0385)	(0.0374)	(0.3267)
Percent lived in same house >= 5 yrs	-0.0309***	-0.0271***	-0.0471
	(0.0046)	(0.0049)	(0.0433)
Percent 25 and older lths educ	0.1011***	0.1177***	0.0263
	(0.0059)	(0.0064)	(0.0563)
Percent 25 and older college or more educ	-0.0790***	-0.0676***	0.0039
	(0.0053)	(0.0061)	(0.0452)
Percent of families with own children	-0.0432***	-0.0161	-0.0457
	(0.0099)	(0.0100)	(0.0756)
Percent ages 5-17	-0.1429***	-0.0015	0.5877***
	(0.0200)	(0.0216)	(0.1700)
Percent ages 65+	-0.3845***	-0.2857***	-0.4288***
	(0.0144)	(0.0149)	(0.1262)
Median age	-0.0321*	0.0758***	0.3493**
	(0.0172)	(0.0187)	(0.1422)
Percent white, NH	-0.0618***	-0.0631***	0.1794***
	(0.0053)	(0.0055)	(0.0385)
Percent black, NH	0.0405***	0.0138**	0.2269***
	(0.0060)	(0.0064)	(0.0458)
Percent Hispanic	-0.0545***	-0.0692***	0.0467
	(0.0055)	(0.0056)	(0.0382)
Percent males	0.1894***	0.1822***	0.7070***
	(0.0222)	(0.0219)	(0.1473)

VARIABLES	No Social Capital Regressors (1)	All Social Capital Regressors, no interactions (2)	All Social Capital Regressors, with interactions (3)
Population density (# pple per 1000 sq mi)	0.0057	-0.0761***	-0.1446**
	(0.0097)	(0.0097)	(0.0652)
Percent of Pop 16+ in LF	-0.0802***	-0.0433***	0.3038***
	(0.0072)	(0.0075)	(0.0682)
Prob Religiosity is high		0.0026	1.536***
		(0.0053)	(0.3362)
			[-0.035]***
Prob Trust is high		-0.2154***	1.073***
		(0.0073)	(0.3814)
			[-0.125]***
Prob Political Activism is high		0.1334***	2.340***
		(0.0083)	(0.4941)
			[0.162]***
Prob Political Engagement is high		0.0097	-1.441***
		(0.0089)	(0.3983)
			[-0.052]***
Prob Community Involvement is high		-0.0419***	-1.532***
		(0.0091)	(0.5188)
			[0.003]
Prob Sociability is high		0.1956***	0.0776
		(0.0119)	(0.5918)
			[0.110]***
Constant	40.50***	27.74***	-17.34
	(2.002)	(2.020)	(16.36)
Observations	63,929	63,929	63,929
R-squared	0.6725	0.6845	0.7373
MSE	31.74	30.58	25.46

Notes: Robust standard errors in parentheses, marginal effects for the interactive model are in brackets. *** p<0.01, ** p<0.05, * p<0.1. Dependent variable is shares of forms in CT occupied housing not returned by mail.